Nolan Cutler

503-860-5906 | cutlern@oregonstate.edu | github.com/nolcut | nolcut.github.io

Education

Oregon State University Honors College — Corvallis, OR

Honors B.S. in Computer Science — *Sep 2023 – June 2027*

- **GPA:** 3.82
- **Awards**: NSF CISE REU (award #2450241), WUE (merit scholarship), Drucilla Shepard Smith Award
- **Research Interests:** Serverless Computing, LLM Agents, Computer Systems
- **Teaching (ULA):** Principles of Computer System Design (CS499/599)

Experience

Computer Science Intern (SULI) — August 2026 – Present

Pacific Northwest National Laboratory – Mentor: Dr. Nathan Tallent — *Richland, WA*

- Investigating locality-aware HPC serverless for the PORTUS Genesis Mission project.

Research Assistant (HiPCastor) — January 2025 – Present

Oregon State University – Mentor: Professor Renato Figueiredo — *Corvallis, OR*

- Core contributor to FaaSr, a middleware enabling decentralized workflow coordination across federated serverless endpoints.
- Designed and implemented CURATE, a multi-agent system for the generation, reuse, and deployment of computational workflows.
- Assisted wildlife conservation efforts with the University of Osaka and Aomori University in Japan; ecological forecasting with Virginia Tech; and smart farming with the agriculture department at OSU.

Publications

[1] **N. Cutler**, C.-C. Kuo, N. Velugoti, K. Newhart, and R. J. Figueiredo, "CURATE: Leveraging LLM agents to compose, catalog, and deploy reproducible workflows," in *Proc. SC Workshops '26: Workshop on Workflows in Support of Large-Scale Science (WORKS)*, Chicago, IL, USA, Nov. 2026, to be published.

[2] A. T. Ramrakhiani, **N. Cutler**, C. C. Carey, A. D. Delany, I. Lerman, R. Q. Thomas, O. Wickman, and R. J. Figueiredo, "EphFlow: Cross-platform FaaS workflows with serverless ephemeral VM orchestration," in *Proc. IEEE Int. Conf. e-Science (eScience)*, 2026.

[3] W. Watanakeesuntorn, S. Date, **N. Cutler**, R. J. Figueiredo, A. Okuyama, and S. Shimojo, "A low-power IoT architecture for environmental monitoring in resource-constrained areas," in *CENTRA (CENTRA 9)*, 2026.

Service

EPICS Kenya — May 2026

Ira A. Fulton Schools of Engineering, Arizona State University — *Western Kenya*

- Helped deliver and install an offline digital library at a rural school and a deaf orphanage in Kenya, providing access to 300+ educational resources in Swahili and English.
- Debugged a makeshift geophone elephant detector in the field by identifying an FFT signal-filtering bug and a discrepancy between the soldered circuit and the circuit specification, enabling the team to present the device to a rural Maasai Mara community as a means of preventing human-wildlife conflict.

Projects

RISC-V Processor and CLI Simulator | *Computer Systems — Python — Docker*

- Designed a single-cycle, RV32I RISC-V processor in SystemVerilog, supporting 33/37 base instructions.
- Developed a visual command-line simulator in Python for running assembly programs, serving as a useful tool for learners to introspect state changes in a single-cycle processor.
- The processor and associated Docker image were adopted into Oregon State University curriculum for Fall 2025 Computer Architecture (undergraduate and graduate) to teach students the fundamentals of datapath design. The Docker image currently has 350 downloads.

CURATE (multi-agent system for computational workflows) | *Python — LLM Agents — Claude SDK*

- Designed a novel multi-agent system that uses LLM agents to manage and develop computational workflows across their entire lifecycle, from generation of functions, to reuse, to automated serverless deployment.
- Implemented 6700 line prototype in Python built on LangGraph and the Claude Agent SDK.
- For more details, please refer to the paper.